

SEPT-ILES, QC, Canada

WMO: 718110

Lat: 50.2181N

Lon: 66.2414W

Elev: 55

StdP: 100.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.9	(c) -25.1	(d) -34.3	(e) 0.1	(f) -27.2	(g) -31.7	(h) 0.2	(i) -24.5	(j) 13.1	(k) -6.1	(l) 11.1	(m) -7.5	(n) 2.8	(o) 300	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	23.0	16.5	21.4	15.7	20.1	15.2	18.1	20.9	17.2	19.6	16.4	18.7	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.2	19.2	16.2	11.6	18.3	15.4	11.0	17.5	51.6	21.0	48.7	19.7	46.2	18.7	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.2	8.1	-33.0	26.8	2.8	1.6	-35.0	27.9	-36.6	28.9	-38.1	29.8	-40.1	30.9
			-33.2	20.0	2.7	1.0	-35.1	20.6	-36.7	21.2	-38.2	21.7	-40.2	22.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-13.8	-12.6	-6.9	0.1	6.2	11.9	15.4	14.9	10.2	4.5	-1.9	-9.2
	DBStd	11.05	6.66	6.54	5.97	3.81	3.09	3.18	2.38	2.54	3.18	3.24	4.74	6.11
	HDD10.0	3492	738	634	523	297	125	16	0	1	35	172	357	594
	HDD18.3	6100	997	867	782	547	377	195	96	109	243	428	607	852
	CDD10.0	443	0	0	0	0	7	72	167	153	42	2	0	0
	CDD18.3	8	0	0	0	0	0	1	4	3	0	0	0	0
	CDH23.3	38	0	0	0	0	1	7	18	12	1	0	0	0
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0
Wind	WSAvg	3.5	3.8	3.8	4.0	4.1	3.6	3.5	3.1	2.9	3.1	3.5	3.6	3.8
Precipitation	PrecAvg	1173	91	77	87	92	94	96	101	99	114	108	109	110
	PrecMax	1545	206	189	219	264	181	199	180	229	233	215	239	233
	PrecMin	908	0	8	21	10	18	34	41	40	35	21	38	39
	PrecStd	176	41	43	43	55	42	38	33	46	43	45	42	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	4.1	6.7	11.7	19.9	24.2	25.5	24.8	21.7	16.4	10.0	5.1
		MCWB	1.4	2.8	2.8	5.6	11.6	15.4	17.2	17.6	16.1	11.8	8.7	4.2
	2%	DB	0.6	1.5	4.0	8.8	16.1	21.4	23.1	22.7	19.0	12.9	7.2	3.1
		MCWB	-0.1	0.2	1.4	4.2	10.1	14.7	16.7	16.7	14.0	9.5	5.7	2.1
	5%	DB	-1.3	-0.3	2.2	6.8	13.7	19.6	21.5	21.0	17.1	11.1	5.8	1.4
		MCWB	-2.0	-1.3	0.1	3.0	8.6	13.7	16.1	15.9	13.2	8.6	4.6	0.6
	10%	DB	-3.4	-2.5	0.9	5.2	11.8	17.7	20.1	19.6	15.5	9.6	4.5	0.0
		MCWB	-4.3	-3.6	-0.7	2.3	7.9	12.8	15.6	15.6	12.5	7.6	3.3	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.0	3.7	6.4	13.2	18.0	19.5	19.2	17.2	12.5	9.1	4.6
		MCDB	2.5	3.8	5.9	9.6	18.0	21.1	22.9	22.4	20.2	14.5	9.6	4.8
	2%	WB	0.0	0.5	1.8	5.0	10.9	16.0	18.2	18.1	15.3	10.6	6.3	2.3
		MCDB	0.6	1.2	3.3	7.9	14.9	19.4	21.0	20.8	17.5	12.3	6.8	2.9
	5%	WB	-1.9	-1.4	0.6	3.8	9.6	14.8	17.3	17.2	14.1	9.2	4.8	0.7
		MCDB	-1.3	-0.3	1.8	6.1	12.6	18.1	19.8	19.5	16.0	10.4	5.5	1.3
	10%	WB	-4.3	-3.5	-0.5	2.7	8.3	13.6	16.5	16.4	13.0	8.0	3.4	-0.6
		MCDB	-3.5	-2.6	0.7	4.7	11.1	16.4	18.8	18.5	14.9	9.1	4.3	0.0
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.5	9.5	7.4	8.5	8.8	8.3	8.4	8.7	7.4	6.9	8.3
		MCDBR	9.8	8.0	8.1	9.7	12.5	12.6	11.0	10.8	10.9	9.9	6.5	6.8
		MCWBR	9.7	7.5	6.4	6.3	7.9	7.2	6.2	6.3	7.2	7.1	5.8	6.8
	5% WB	MCDBR	9.8	7.6	7.3	8.1	11.0	10.5	8.7	8.5	9.2	7.9	6.0	6.9
		MCWBR	9.8	7.3	6.3	5.8	7.5	6.8	5.7	5.7	7.0	6.5	6.0	7.1
Clear-Sky Solar Irradiance	taub	0.262	0.268	0.293	0.342	0.365	0.391	0.425	0.386	0.363	0.327	0.294	0.267	
	taud	2.363	2.323	2.323	2.308	2.393	2.369	2.306	2.418	2.454	2.525	2.504	2.393	
	Ebn at Noon	793	882	917	901	892	867	829	848	832	806	754	730	
	Edh at Noon	63	84	102	117	113	117	123	104	91	71	56	53	
All-Sky Solar Radiation	RadAvg	1.11	2.13	3.44	4.52	5.21	5.50	5.16	4.37	3.29	1.98	1.05	0.81	
	RadStd	0.13	0.19	0.28	0.37	0.33	0.47	0.51	0.27	0.23	0.17	0.10	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.46	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	+1.25	N/A	+0.26	+0.34	N/A	N/A	N/A	N/A

Nomenclature: See separate page